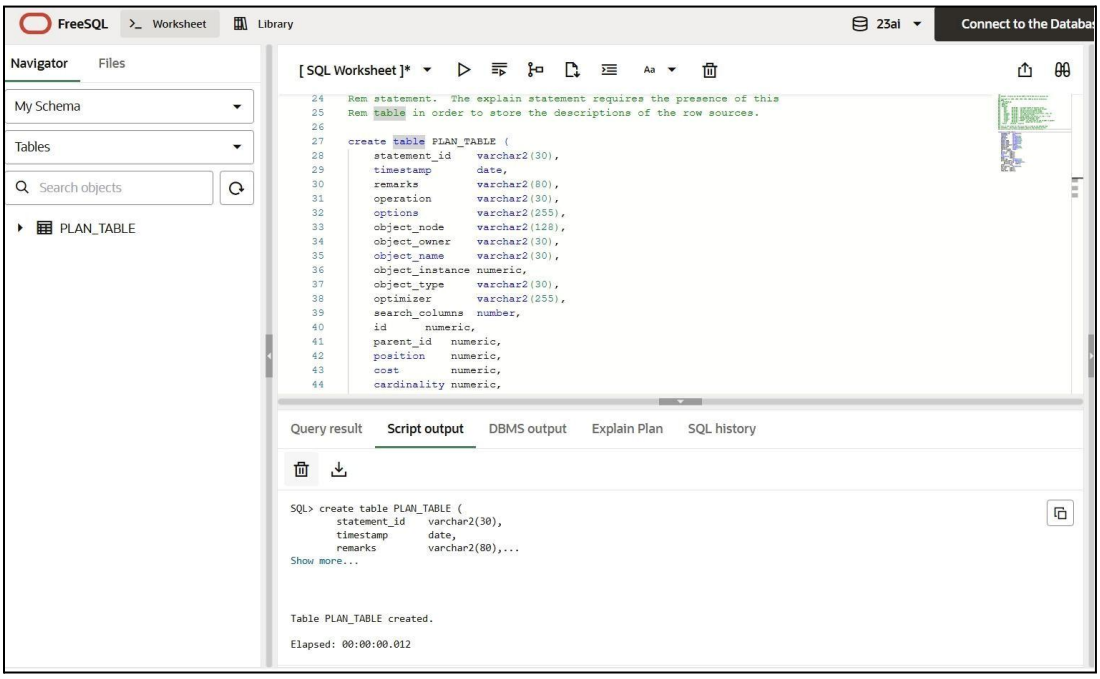
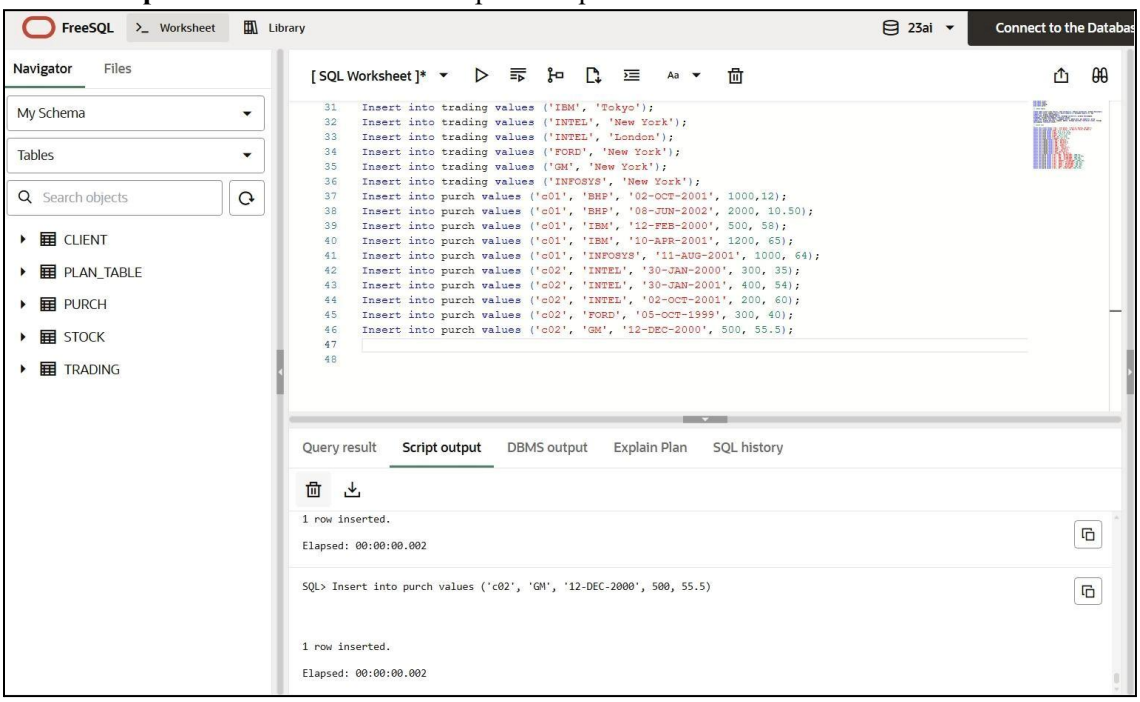
Create the PLAN_TABLE - utlxplan.sql



Create the sample database schema - SampleDB.sql

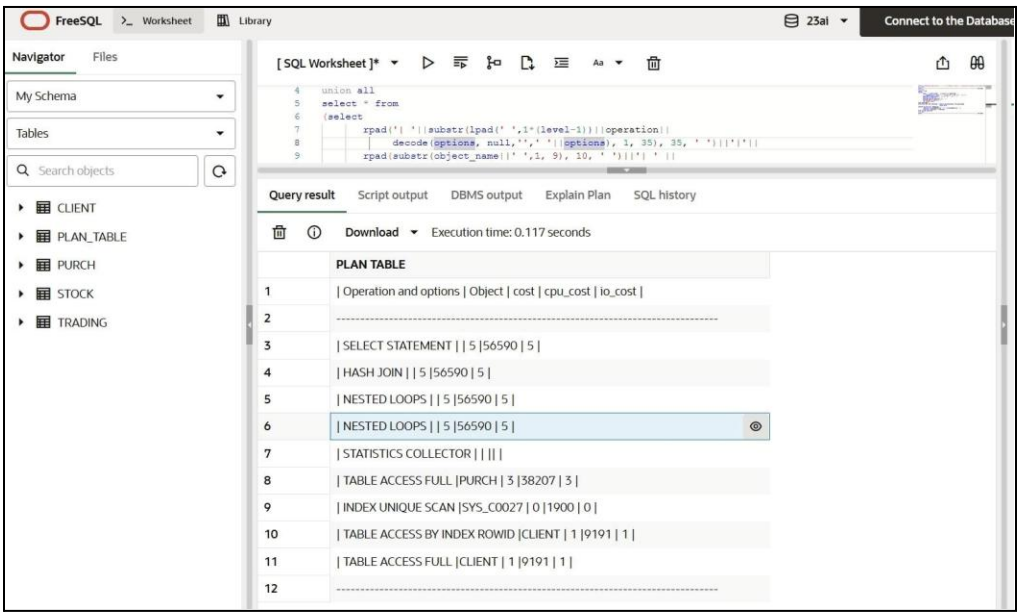


a) Inserting the execution plan into PLAN_TABLE.

```
ALTER SESSION SET OPTIMIZER_MODE = ALL_ROWS;
ALTER SESSION SET "_optimizer_cost_model" = CPU;

EXPLAIN PLAN FOR
SELECT c.clno, c.name
FROM client c, purch p
WHERE c.clno = p.clno AND p.qty > 1000;
```

b) Run Oracle's script utlxpls.sql to display the plan.



Query Plan Explanation :

- SELECT STATEMENT – overall query execution.
- HASH JOIN / NESTED LOOPS – used to join CLIENT and PURCH on clno.
- TABLE ACCESS FULL (PURCH) – full scan of Purch table to filter qty > 1000.
- INDEX UNIQUE SCAN (SYS_C0027) – primary key index on Client used.
- TABLE ACCESS BY INDEX ROWID (CLIENT) – retrieves matching Client rows.
- TABLE ACCESS FULL (CLIENT) – an additional full scan of Client table.

Costs:

- COST = 5, CPU_COST = 56,590, IO_COST = 5.
- The high CPU cost shows the query is more CPU-intensive than I/O-intensive.

c) Create Indexes :

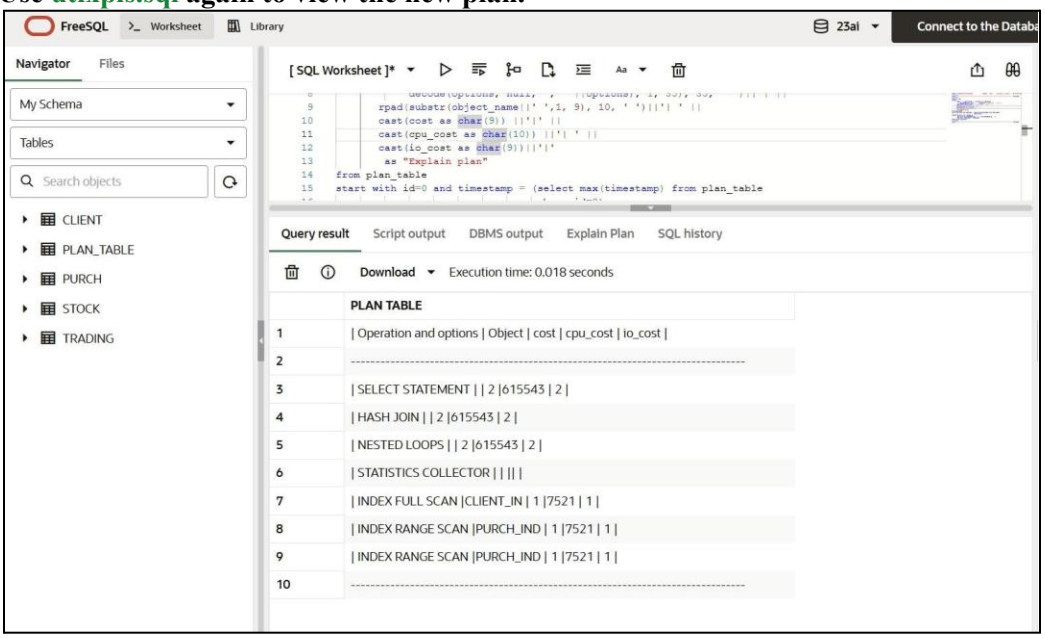
CREATE INDEX purch_index ON purch(qty, clno);
CREATE INDEX client_index ON client(clno, name);

d) Re-run the same EXPLAIN PLAN query from part (a).

ALTER SESSION SET OPTIMIZER_MODE = ALL_ROWS;
ALTER SESSION SET "_optimizer_cost_model" = CPU;

EXPLAIN PLAN FOR
SELECT c.clno, c.name
FROM client c, purch p
WHERE c.clno = p.clno AND p.qty > 1000;

Use utlxpls.sql again to view the new plan.



Query Plan Explanation :

After creating indexes on PURCH(qty, clno) and CLIENT(clno, name), the execution plan changed:
- SELECT STATEMENT – overall query execution.
- HASH JOIN / NESTED LOOPS – join between CLIENT and PURCH on clno.
- INDEX FULL SCAN (CLIENT_IND) – Oracle scans the Client index instead of a full table scan.
- INDEX RANGE SCAN (PURCH_IND) – Oracle uses the Purch index to filter rows where qty > 1000.
- INDEX RANGE SCAN (PURCH_IND) – repeated for join condition.

Costs:

- COST = 2, CPU_COST = 615,543, IO_COST = 2.
- Compared to part (b), the overall cost is lower, showing improved efficiency.